

ABHISHEK SHANKAR

774-519-0926 | Mountain View, CA | abhishekshankar88@gmail.com | linkedin.com/in/abhishekshankar97

Work Experience

RhythmX AI

AI Engineer Consultant

December 2024 – Present

Palo Alto, CA

- Built and deployed production GenAI recommendation agents on Azure, expanding clinical AI coverage from 4 to 200+ medical conditions at 98%+ accuracy
- Engineered LLM evaluation frameworks using LLM-as-a-judge and human-in-the-loop clinical feedback to benchmark model precision, recall, and faithfulness across AI features; established Go/No-Go accuracy thresholds for production deployments
- Built Batch API inference pipelines via Temporal, processing 200,000+ requests, reducing API overhead by \$4,000 weekly; profiled MongoDB queries and system architecture to cut end-to-end inference latency by 50%
- Operated in HIPAA-compliant regulated environments enforcing data security across HL7 v2 and FHIR pipelines and AI inference workflows; developed smoke and regression test suites for STAGING and PROD deployments in compliance-gated release cycles
- Designed RESTful APIs serving LLM predictions to clinical applications and built CI/CD pipelines on Azure (App Services, Container Apps, AKS) for Dockerised AI services, integrating AI capabilities seamlessly into enterprise workflows

Boehringer Ingelheim

Data Scientist Co-op

July 2023 – December 2023

Ridgefield, CT

- Built GraphRAG knowledge extraction pipeline using Stardog and LLMs to automate NLP-driven analysis of 1,000+ supply chain PDFs, enabling structured entity extraction, text classification, and reporting at scale
- Delivered real-time analytics dashboards for 10+ drug launches using AWS CDK and Databricks, embedding ML model outputs into enterprise decision workflows for commercial teams
- Created 25+ Python and PySpark tutorial notebooks standardising ML development practices and accelerating enterprise Databricks adoption across the data science team

Adqvest Capital Pvt. Ltd.

Data & Software Engineer

April 2020 – June 2022

Chennai, India

- Built NLP sentiment classification pipelines using SpaCy and PageRank algorithms for automated text analysis, named entity recognition, and report generation, retaining 5+ clients worth \$2M+ AUM
- Deployed 100+ Python scrapers and AWS RDS infrastructure to process gigabytes of structured and unstructured data for downstream ML workflows, powering real-time market analysis for institutional clients

Projects

SupplyChainInsights – GraphRAG & LLM Fine-Tuning Platform ([GitHub](#) & [Medium](#)) May 2024 – August 2024

Python, Neo4j, LangChain, Hugging Face Transformers, OpenAI, PyTorch, RAGAS, Streamlit

- Featured in Neo4j's *This Week in Neo4j* newsletter – built a RAG platform modelling 1,000+ ARV and HIV supply chain entities with Cypher-driven traversal; fine-tuned Llama 3 and Phi-3 using Hugging Face Transformers for domain-specific NL-to-Cypher query conversion
- Evaluated model quality using RAGAS framework measuring Faithfulness and Answer Relevancy; applied Graph Data Science similarity algorithms with OpenAI vector search to surface entity-level recommendations from structured datasets

Parallel Image Captioning with PyTorch DDP ([GitHub](#))

Python, PyTorch, CUDA, Multi-GPU

January – May 2024

- Implemented distributed data-parallel training with PyTorch DDP achieving 3x faster multi-GPU training; built encoder-decoder image captioning architecture with mixed precision training and parallel data loading pipelines

Education

Northeastern University

Master of Science in Information Systems (Data Science)

August 2024

Boston, MA

SRM Institute of Science and Technology

Bachelor of Technology in Mechanical Engineering

May 2019

Tamil Nadu, India

Skills

LLM & GenAI: Fine-tuning (Llama 3, Phi-3, GPT-4), RAG (Vector & Graph), Prompt Engineering, LLM Evaluation, LangChain, Hugging Face Transformers, OpenAI

Programming & ML: Python (Expert), PyTorch, TensorFlow, Scikit-Learn, SQL, JavaScript, PySpark

Cloud & ML Platforms: Azure ML, AWS (CDK, SageMaker, RDS), GCP, Databricks, Snowflake

Databases: Neo4j, PostgreSQL, MongoDB, Stardog (Knowledge Graphs), ChromaDB (Vector)

DevOps & Tools: RESTful APIs, CI/CD, Docker, Git, Streamlit, Airflow, Kubernetes (AKS), Tableau